

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

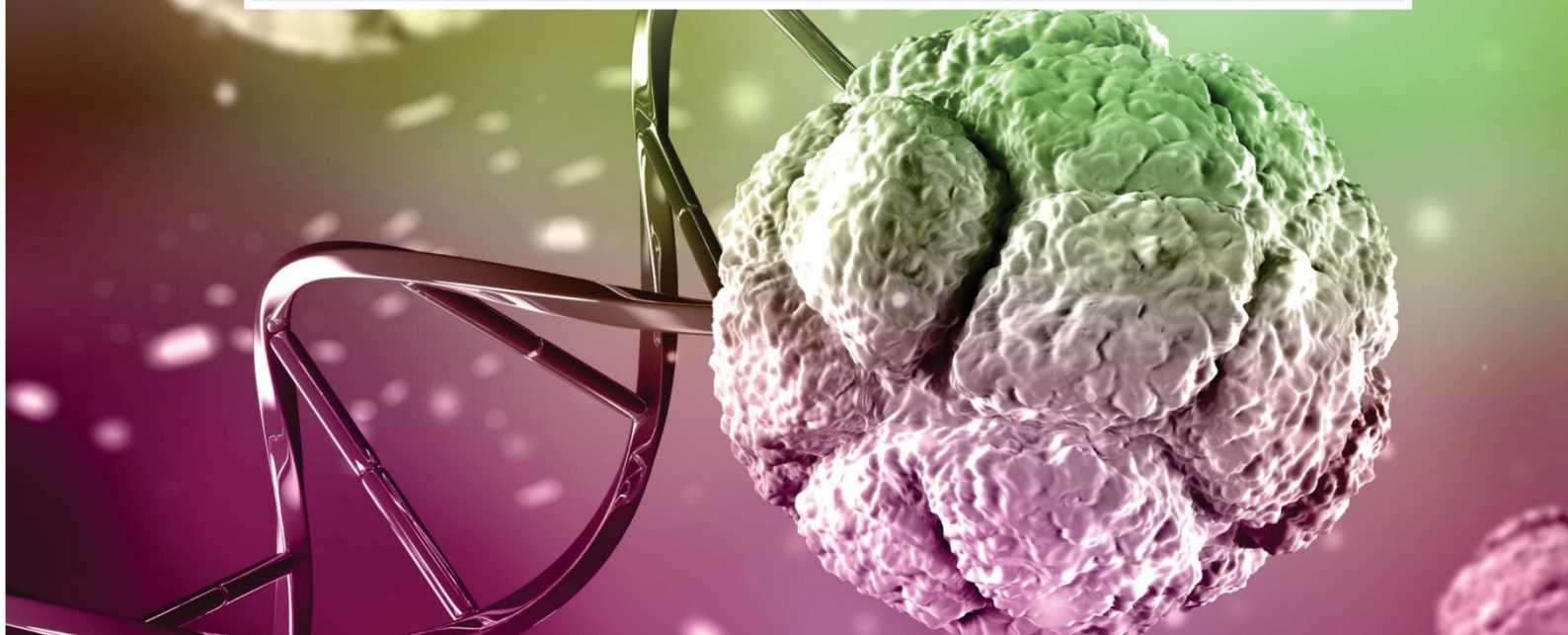
BIOLOGY

Topic wise
Notes

LECTURE

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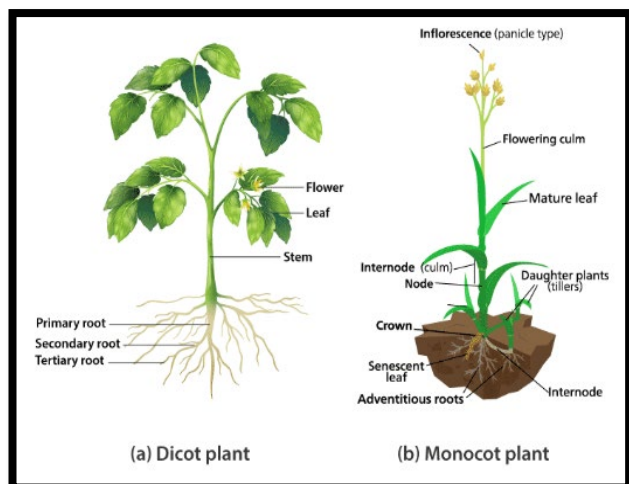
Plant Morphology



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Plant Morphology

Plant morphology is the study of the external form, structure, and development of plants, focusing on vegetative organs (roots, stems, leaves) and reproductive organs (flowers, fruits, seeds).



Plant is made up of three basic structures:

- (i) **Root:** Root originates from radical of seed. primary root grows further to form tap root when primary root is replaced by thread like Structure known as adventitious/ fibrous root.
 - +ve Geotropic — Growth towards gravity
 - +ve Hydrotropic— Towards underground water.
 - -ve phototropic— opposite to sunlight

Root modifications:

- **Storage Roots:** Swollen roots that store food.
 - **Example:** Carrot, Casava
- **Prop Roots:** Adventitious roots that grow from the stem to provide support.
 - **Example:** Banyan tree
- **Respiratory Roots (Pneumatophores):** Roots that grow upward from the soil in waterlogged environments to facilitate gas exchange.
 - **Example:** Rhizophora

Type of Roots:

1. Tap Root System

- Develops from radicle

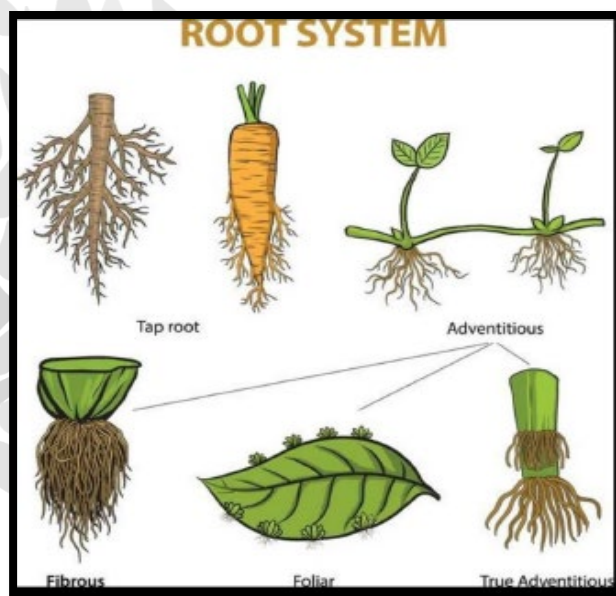
- Main root with lateral branches
- Example: Mustard, Carrot

2. Fibrous Root System

- No main root, many thin roots
- Example: Wheat, Rice

3. Adventitious Roots

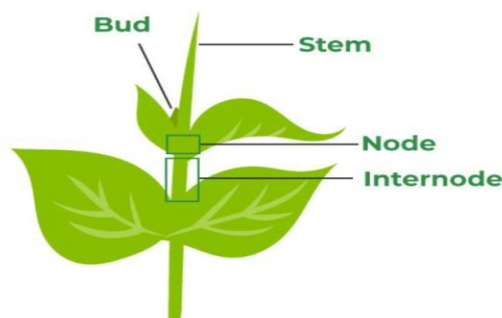
- Arise from stem/leaves
- Example: Banyan (prop roots), Maize



STEM

- Part of plant developed from plumule of seed Node during germination.
- Contains nodes and internodes.
- Those region on stem from where leaves are born known as node.

Structure of Stem



Stem Modification:

- **For Storage:** Underground stems that store food and act as storage organs.





- **Example:** Rhizomes (ginger, turmeric), corms (Colocasia), bulbs (onion)
- **For Photosynthesis:** Modified stems that are green and flattened to carry out photosynthesis.
 - **Example:** Cactus (cladodes), Asparagus (cladodes)
- **For Vegetative Propagation:** Stems grow partially above and partially below the ground and can produce new plants from their nodes.
 - **Example:** Runner (strawberry), Stolon (Bermuda grass)

Sub-aerial modification of stem

- A - sub-aerial modification of stem These are creeping stems.
- **Runner** - In these stems roots develop at lower side and leaves at upper side from the nodes **Eg.** Cynodondactylon (doob grass), Oxalis, Strawberry (Fragaria ananassa).
- **Stolon** - In it branches develop from the lower part of the main stem and grow for some distance-like arch and finally touch the ground to give rise to new shoot. **Eg.** Fragaria Vesca (Wild strawberry), jasmine, peppermint
- **Sucke** - In it the main stem grows in the soil but branches develop from (basal and underground nodes and comes out from the soil. **Eg.** Mint, pineapple, Chrysanthemum, banana.
- **Offset-** Generally these occur in aquatic plants which have fragile stems. Internodes of offset are small & thicker and each node bears a rosette of leaves and a tuft of roots. It is also known as aquatic runner. **Eg.** Pistia, Aquati Eichhornia

Underground stem modification:

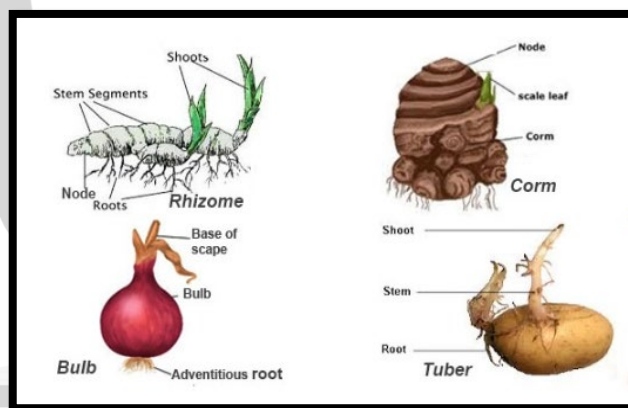
Underground stem modifications are stems that grow below the soil surface and are modified to perform special functions such as storage of food, perennation (survival), and vegetative reproduction.

- **Rhizom**→ Ginger, Garlic, Banana

- **Tuber**→ potato
- **Corm**→ Zaminkand, Colocasia, Saffron, Alocasia
- **Bulb**→ Onion, Garlic

Aerial modification of stem:

- **Stem Tendril-** In this type of modification axillary bud forms tendril in place of branches and helps in climbing of those plants which have weak stem. **E.g.** Grapes/Grapevines, Passiflora, gourds (cucumber, pumpkins, watermelon)
- **Stem thorn** - Thorn develops usually from axillary bud of the stem.
 - It may bear leaves and flowers. **E.g.** Carissa (karonda), Bougainvillea, pomegranate, Citrus.
 - Thorn protects plants from browsing animals.
 - It is a woody pointed structure.
 - Phylloclade Stem is modified into a fleshy flat (Opuntia) or fleshy cylindrical (Euphorbia) and green leaf.
 - The leaves are modified into spines **E.g.** Opuntia, Euphorbia, cactus, Casuarina (cylindrica).



LEAF

A leaf is a flattened, green, lateral appendage of the stem, borne at a node, and is the main organ of photosynthesis in plants.

Leaf modifications:

- **For Support:** Modified leaves that are thin and thread-like, helping the plant climb.
 - **Example:** Tendrils on a pea plant

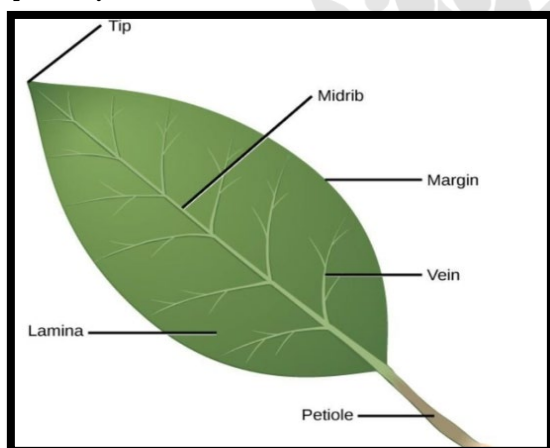




- **For Defense:** Modified leaves that are sharp and pointed to protect the plant from herbivores.
 - **Example:** Spines on a cactus
- The leaves develop from the nodes. Their main function is to carry out photosynthesis or food formation, axillary buds are found in the axil of leaves.

Part of leaf:

- Leaf base**– part of leaf from which it is attached to stem known as leaf base.
- Petiole**– Part of leaf through which lamina is connected to stem leaf base.
- Lamina**– Exposed surface of leaf, broad surface area mainly responsible for photosynthesis.



- Stipule**– Two lateral appendages either side of leaf base known as stipule.

MODIFICATION OF LEAVES:

- When leaves are modified into different structures then it is called modification of leaves.
- **Leaf tendril** - In some plants whole leaf is modified into a wirelike structure which is called leaf tendril.
- Tendril helps in climbing. Eg. Lathyrus aphaca (wild pea)
- **Leaf spine** - Leaves are modified into pointed spines. Eg. Opuntia, Cacti, Argemone.
- **Leaf pitcher**– Leaves of some plants are modified into pitcher shaped structure. Eg. Nepenthes (pitcher plant) (Only lamina is modified into pitcher).
- Water is stored in the pitcher of Dischidia (complete leaf is modified into pitcher).
- In Nepenthes insectivorous pitcher while in Dischidia - non insectivorous pitcher is formed.

- **Leaf bladder** - In some plants, leaves are modified into bladder like structure. Eg. Utricularia (bladder wort)
- **Phyllode** - In some plants petiole becomes flat leaf like green, synthesises food and functions as normal leaf. Eg.: Australian acacia, Parkinsonia.
- **Leaflet tendril** - When leaflet is modified into tendril-like structure then it is called leaflet tendril. Eg. Pisum sativum (garden pea), Lathyrus odoratus (sweet pea).
- **Note:** Dionaea (Venus flytrap) is insectivorous plant and it also has modified leaves.
- **Leaflet hook** - Eg. Cat's nail (Bignonia unguis cati)

